

KARTEEK ALLA

(Cloud Engineer/DevOps Engineer)

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PROFESSIONAL SUMMARY:

- Over **9 years of experience** as a **Cloud and DevOps Engineer**, specializing in **cloud infrastructure automation, container orchestration, and CI/CD pipeline development** across **Azure and Google Cloud Platform (GCP)** environments.
- Proven ability to architect, deploy, and manage enterprise cloud solutions with a strong focus on **scalability, security, and cost efficiency** using modern DevOps practices.
- Provisioned and managed **Azure cloud services** across multiple domains, including **Azure VMs, AKS, networking (Virtual Networks (VNet), Subnets, NSGs, Load Balancers, Application Gateways, Azure DNS), storage (Blob Storage, Azure Files), databases (Azure SQL Database), serverless and integration (Azure Functions, Logic Apps, API Management, Service Bus), security and identity (Key Vault, Azure AD, RBAC, ADFS, SSO), and business continuity (Backup, Site Recovery)**; automated provisioning using **Terraform, ARM Templates, and Azure CLI**, and implemented monitoring and observability using **Azure Monitor, Log Analytics, and Application Insights**.
- Experience in **Google Cloud Platform (GCP)** using services such as **Compute Engine, Google Kubernetes Engine (GKE), Anthos, Cloud Run, Cloud Functions, IAM, Cloud Storage, Cloud SQL, VPC Networking, Pub/Sub, Cloud Build, BigQuery, and Google Cloud Service Mesh (Istio on GKE)**.
- Hands-on exposure to **OpenStack** components (**Nova, Neutron, Cinder, Glance, Horizon**) through lab environments and internal **POCs**; experienced in automating **OpenStack deployments** using **Ansible and Terraform**, integrating with Ceph for scalable block/object storage and Neutron for advanced networking (VXLAN, VLAN, L3 HA). Familiar with Cisco UCS integration patterns for bare-metal provisioning and Ironic-based infrastructure lifecycle management.
- Extensive experience in creating cloud Infrastructure from the scratch for projects and maintaining as per organization standards. Strong expertise in **infrastructure as code (IaC)** using **Terraform, CloudFormation** as well as cloud-native application deployment and hybrid/multi-cloud strategies.
- Managed **Identity and Access Management (IAM)** in **Azure AD** and **GCP IAM**, defining **custom roles, RBAC policies, Service Principals, and Managed Identities** to follow least-privilege access.
- Integrated **SSO** using **Active Directory, ADFS**, and third-party identity providers, ensuring secure and centralized authentication across cloud platforms.
- Implemented **Terraform Cloud** for state management and policy enforcement, enabling collaboration across teams in multi-environment deployments.
- Worked on **ARM templates** for declarative provisioning and **automation** of Azure resources, integrated into CI/CD pipelines.
- Strong hands-on experience in **Kubernetes** (AKS and GKE), deploying microservices using **Helm Charts**, setting up **HPA, Pod Disruption Budgets, and Pod Affinity/Anti-affinity rules** for workload distribution.
- Implemented **service mesh** using **Istio** for advanced routing, mTLS, observability, and security across microservices within **Kubernetes**.
- Managed **Kubernetes manifests**, handled multi-tenant clusters, and developed internal tools to streamline deployment and cluster monitoring.
- Designed and built **CI/CD pipelines** using **Azure DevOps, Jenkins, GitHub Actions**, and , automating everything from code builds, security scanning, testing, to multi-cluster deployments.

- **Implemented automated CI/CD pipelines** with integration of build tools, artifact repositories, and features like **Webhooks**, **Triggers**, and **Approvals** to ensure smooth delivery workflows with **rollback capabilities**.
- Automated deployment processes to **Azure AKS**, **GKE**, and **VM-based** infrastructure using **Ansible**, **Helm**, and **container registries**.
- Designed and implemented **Azure cloud networking architectures**, including **Virtual Networks (VNETs)**, **Subnets**, **Network Security Groups (NSGs)**, **VPN Gateways**, **VNet Peering**, and **Private Endpoints** to enable **secure, scalable, and resilient network connectivity** across hybrid and cloud-native environments.
- Configured **GCP VPCs**, **Firewall Rules**, **Peering**, and **Cloud NAT**, implementing **load balancing** and **failover strategies** across zones for high availability.
- Set up observability using **Azure Monitor**, **Log Analytics**, **Application Insights**, and custom **Workbooks** for real-time performance tracking and alerting.
- Deployed and configured **ELK Stack**, **Grafana**, **Prometheus**, **AppDynamics**, and **Splunk** to collect, visualize, and analyze application and infrastructure metrics.
- Diagnosed complex system issues using logs, metrics, and traces, improving uptime and reducing mean time to resolution (MTTR).
- Proficient in scripting and automation using **Python (Boto3)**, **Bash**, and **PowerShell** to orchestrate daily operations, automate deployments, and manage user access and services.
- Automated snapshots, backup policies, and environment configurations using **Lambda**, **Cloud Scheduler**, and custom scripts across cloud platforms.
- Developed and maintained **monitoring dashboards**, wrote custom **alerts**, and implemented **incident response** integrations with Slack and email systems.
- Experience with **Docker** containerization, building custom images, and managing private container registries for secure application packaging.
- Automated build/test processes using containerized environments, ensuring consistent and reliable testing across environments.
- Managed **Git repositories**, implemented branching strategies, pull request workflows, and integrated code quality tools like **SonarQube** into CI pipelines.
- Led cloud migration initiatives from on-prem to **Azure** and **GCP**, using tools like **Azure Migrate**, **Database Migration Service**, and **GCP Migrate for Compute Engine**.
- Collaborated across **Dev**, **QA**, and **Ops** teams to align infrastructure design with business goals and compliance standards.
- Delivered infrastructure improvements that reduced manual effort, improved release cycles, and increased system reliability using **DevOps-first approaches**.

CERTIFICATIONS:

- Microsoft Certified: Azure Data Fundamentals (AZ-900)
- Microsoft Certified: Azure DevOps Engineer Expert (AZ-400)
- Google Cloud Certified: Associate Cloud Engineer

TECHNICAL SKILLS:

Cloud Platforms	Microsoft Azure, Google Cloud Platform (GCP)
Infrastructure as Code (IaC)	Terraform, ARM Templates, Azure CLI, CloudFormation, Azure DevOps Pipelines, Terraform Cloud, pulumi
Containerization	Docker, Docker Compose, Azure Container Registry (ACR)
Container Orchestration	Kubernetes (AKS, GKE, OpenShift), Helm, Istio, Ingress Controllers
CI/CD Tools	Jenkins, GitLab Actions, Azure Pipelines
Configuration Management	Ansible, Chef, Puppet, Azure Automation, YAML, JSON
Scripting Languages	PowerShell, Python(Boto3, Azure SDK), Bash, Shell, Ruby (RSpec), GoLang (basic proficiency for custom OpenStack operator tools)
Version Control	Git, SVN, GitHub, GitLab
Monitoring Tools	Azure Monitor, Azure Log Analytics, GCP Stackdriver, Splunk, ELK Stack (Elasticsearch, Logstash, Kibana), Grafana, AppDynamics, Monit, Filebeat
Networking / Security	Azure VNet, Subnets, NSG, Azure Load Balancer, Application Gateway, Azure Firewall, Azure Front Door, Azure Traffic Manager, VPN, ExpressRoute, DNS (Azure & GCP), SSL/TLS, IAM, Azure RBAC, Conditional Access
Serverless & Messaging	Azure Functions, RabbitMQ, Azure Event Grid
Storage & Databases	Azure Blob Storage, Azure SQL Database, Azure Synapse Analytics, Azure Key Vault, GCP Cloud SQL, DynamoDB, MySQL, PostgreSQL, SQL Server, Azure Data Factory
Identity & Access Management	Azure Active Directory, ADFS, LDAP, Azure AD SSO, Federation Services, Group Policies
DevOps Practices	DevSecOps, Blue-Green Deployments, Zero-Downtime Deployments, Change & Incident Management, Agile
Build Tools	Maven, ANT, Griddle

Professional Experience:

Hybrid Cloud Engineer,
Indiana University Health

Jan 2024 - Present

Project Description:

As a Cloud Engineer, I Designed and managed scalable **cloud infrastructure** on **Azure** and **GCP**, automating provisioning with **ARM templates**, **Terraform**, and **CLI tools**. Containerized applications using **Docker**, **AKS**, and **GKE** with **Istio**. Built **CI/CD pipelines** with **Jenkins** and implemented **monitoring** using native cloud tools and third-party platforms. Automated workflows using **scripting** and **configuration management**.

Key Responsibilities:

- Designed and deployed scalable **Azure infrastructure** using **ARM templates**, **Terraform**, and **Azure CLI**, provisioning core services such as **Azure VMs**, **AKS clusters**, **Azure Blob Storage**, **Azure Functions**, **Application Gateway**, **Load Balancer**, **Azure DNS**, **Backup Vault**, **Site Recovery**, **Active Directory (AD)**, **ADFS**, **RBAC**, and **Azure Monitor** to support production-grade workloads.
- **Managed Google Cloud Platform (GCP) resources** including **Compute Engine**, **Google Kubernetes Engine (GKE)** with autoscaling and node pools, **Cloud Storage**, **Cloud Functions**, **Cloud SQL**, **Pub/Sub**, and configured networking components such as **VPCs**, **Firewall Rules**, **Cloud NAT**, and **VPC Peering**; automated provisioning via **Terraform**, **gcloud CLI**, and **Deployment Manager**.
- Built and maintained robust **CI/CD pipelines** with **Jenkins**, integrating **Git** for source control, enabling automated build, test, and deployment processes—including container image builds and multi-cluster Kubernetes deployments—with rollback support for reliability.
- **Orchestrated microservices deployments and traffic management** on **Kubernetes clusters** (AKS and GKE), leveraging **Istio service mesh** for secure service-to-service communication, observability, traffic routing, and policy enforcement.
- Led a **PoC for OpenStack** (Victoria) on Ubuntu, automating deployment via **Ansible** and **Terraform** with **Ceph RBD storage** and **Neutron VXLAN** networking.
- Integrated Cisco UCS via **Ironic** for bare-metal provisioning alongside VM workloads; used **Glance** and **Horizon** for image management and tenant self-service.
- **Designed and managed cloud networking and security architectures** in Azure and GCP, including **Virtual Networks (VNETs)**, **Subnets**, **Network Security Groups (NSGs)**, **VPN Gateways**, **VNet Peering**, **Private Endpoints** in Azure, and **VPCs**, **Firewall Rules**, and **Cloud NAT** in GCP to ensure secure, scalable connectivity.
- **Administered Identity and Access Management (IAM)** across **Azure Active Directory (Azure AD)** and **GCP IAM**, configuring **Role-Based Access Control (RBAC)**, **Service Principals**, **Custom Roles**, **ADFS**, and **Single Sign-On (SSO)** to enforce least-privilege security principles.
- **Implemented comprehensive monitoring, logging, and alerting solutions** using **Azure Monitor**, **Log Analytics**, **Application Insights**, **GCP Stackdriver**, **ELK Stack** (**Elasticsearch**, **Logstash**, **Kibana**), **Logstash**, **Filebeat**, **Grafana**, and **Splunk** to enable real-time observability, proactive issue detection, and troubleshooting.
- **Automated infrastructure provisioning, configuration management, and operational workflows** with **Ansible**, **PowerShell**, **Python** (using **Azure SDK** and **GCP client libraries**), and **Bash scripting** to improve deployment speed, consistency, and environment reliability.
- **Managed container registries and artifact repositories** such as **Azure Container Registry (ACR)** and integrated asynchronous messaging with **RabbitMQ** to enable scalable, fault-tolerant microservices architectures.
- **Performed performance tuning and root cause analysis** of cloud infrastructure using custom dashboards and visualizations in **Grafana** and **Kibana**, complemented by application performance monitoring tools like **AppDynamics** and system monitoring with **Monit** to ensure SLA adherence.
- **Led change management, incident response, and release management** processes to maintain high availability, reduce downtime, and support seamless deployment of critical production workloads.
- **Collaborated cross-functionally** with development, QA, security, and operations teams to align cloud infrastructure solutions with business goals and compliance standards.
- **Conducted cloud migration assessments and modernization initiatives** by leveraging **Azure Migrate** and **GCP migration tools**, facilitating smooth transitions from on-premises to cloud environments.
- **Developed custom automation scripts and tools** to streamline repetitive operational tasks, including snapshot scheduling, backup automation, and user provisioning.

- **Maintained documentation and runbooks** for infrastructure architecture, deployment procedures, and incident handling to support knowledge sharing and onboarding.

Environment & Tools:

Azure VMs, AKS, ACR, Blob Storage, Load Balancer, Application Gateway, DNS, Functions, Monitor, Active Directory, ADFS, RBAC, Backup Vault, Site Recovery, Azure CLI, ARM Templates, Azure DevOps, Compute Engine, GKE, Cloud Storage, Cloud Functions, Cloud SQL, Pub/Sub, VPC, Firewall Rules, Cloud NAT, VPC Peering, IAM, gcloud CLI, Deployment Manager, Jenkins, Git, Docker, Helm, Istio, VNets, Subnets, NSGs, VPN Gateways, VNet Peering, Private Endpoints, IAM - Azure AD & GCP IAM, RBAC, SSO, ADFS, Azure Monitor, Log Analytics, Application Insights, Stackdriver, ELK Stack, Grafana, Splunk, Logstash, Filebeat, AppDynamics, Monit, Ansible, PowerShell, Python - Azure SDK & GCP client libraries, Bash, RabbitMQ, SVN, Linux - RHEL, CentOS, Jira, Confluence, Nexus, Artifactory.

DevOps Engineer,
AT&T, Tech Mahindra, Hyd, IND.

Jun 2020 – Aug 2023

Project Description:

Engineered and maintained scalable and secure **cloud infrastructure** on **Microsoft Azure**, automating deployments, orchestrating containers, and integrating comprehensive monitoring solutions. Delivered **CI/CD pipelines** and **infrastructure automation** while enforcing security policies and operational best practices.

Key Responsibilities:

- Designed and provisioned **Azure infrastructure** using **ARM templates**, **Terraform**, and **Azure CLI**, including **Virtual Machines (VMs)**, **Virtual Networks (VNets)**, **Subnets**, **Network Security Groups (NSGs)**, **Load Balancers**, **Application Gateways**, **Traffic Manager**, and **DNS** for secure and highly available environments.
- Built and managed **Azure Kubernetes Service (AKS)** clusters with multi-node pools, implemented **Helm** charts for microservices deployment, and configured **Istio service mesh** for traffic routing, security, and observability.
- Developed, maintained, and optimized **CI/CD pipelines** using **Jenkins** and **Azure DevOps Pipelines**, integrating automated builds, tests, container image creation, and multi-environment deployments with rollback capabilities.
- Automated configuration management and environment provisioning with **Ansible**, leveraging **PowerShell** and **Python** scripting to streamline operations, patch management, and service deployments.
- Configured and optimized **Azure Storage** solutions including **Blob Storage**, **Azure SQL Database**, and **Azure Key Vault** to secure data and manage secrets efficiently.
- Implemented serverless functions using **Azure Functions** for automation tasks such as scheduled backups, snapshot management, and operational alerts.
- Designed and deployed **Azure Data Factory** pipelines to facilitate seamless data migration and ETL processes between on-premises and cloud data stores.
- Managed identity and access through **Azure Active Directory (AAD)**, configuring **Role-Based Access Control (RBAC)**, **Conditional Access Policies**, **Single Sign-On (SSO)**, and **Active Directory Federation Services (ADFS)** to ensure secure, least-privilege access.

- Implemented monitoring and alerting frameworks with **Azure Monitor**, **Log Analytics**, and **Application Insights**, reducing incident response times and improving system reliability.
- Built comprehensive dashboards and visualizations in **Grafana** and **Kibana**, correlating logs and metrics from **ELK Stack** and other monitoring tools for proactive performance management.
- Deployed and tuned **ELK Stack (Elasticsearch, Logstash, Kibana)** for centralized log collection and analysis of applications and infrastructure.
- Administered **Azure Site Recovery (ASR)** and **Azure Backup Vaults** to enforce disaster recovery and business continuity strategies.
- Managed DNS configurations and custom domains using **Azure DNS** to ensure reliable and compliant domain management.
- Implemented advanced deployment strategies such as **blue-green** and **canary deployments** using **Azure App Services** and **AKS** for zero-downtime releases.
- Led incident response and change management processes, collaborating with cross-functional teams to minimize downtime and improve service availability.
- Integrated security tools and vulnerability scanners into CI/CD workflows to enforce **DevSecOps** best practices.
- Automated multi-region and multi-environment resource provisioning using **Terraform Cloud** and pipeline-driven deployments for scalability and fault tolerance.
- Optimized cloud costs and resource usage through continuous monitoring, scaling policies, and right-sizing of Azure resources.
- Documented infrastructure architectures, deployment processes, and operational runbooks to ensure knowledge sharing and compliance.

Environment & Tools:

Azure VMs, VNets, NSGs, AKS, ACR, Azure SQL, Azure Monitor, Log Analytics, App Services, Traffic Manager, Load Balancer, Front Door, Key Vault, Data Factory, ARM Templates, DNS, CDN, Site Recovery, Automation Runbooks, Synapse, Terraform, Ansible, Jenkins, GitLab CI/CD, Docker, Kubernetes, Helm, Istio, PowerShell, Python, Azure DevOps Pipelines, ELK Stack, Splunk, Grafana, Kibana, Artifactory, YAML, JSON

DevOps Engineer

Virtusa Consulting Services Private Limited, IND,

Dec 2018 – May 2020

Project Description:

Led end-to-end cloud migration initiatives, transitioning enterprise applications and infrastructure from on-premises to **Microsoft Azure**. Streamlined **infrastructure as code (IaC)** using **ARM templates**, **Terraform**, and **Azure CLI** to ensure reproducibility and scalability. Built and managed **CI/CD pipelines** using **Jenkins** and handled the orchestration of **containerized workloads** with **Docker**, **Kubernetes**, and **Azure Kubernetes Service (AKS)**. Enhanced observability with **monitoring and log analytics tools** such as **Grafana**, **Prometheus**, **ELK Stack**, and **Splunk** while automating routine tasks through **Python scripting**.

Key Responsibilities:

- Designed and automated scalable Azure infrastructure using **Terraform**, **ARM templates**, and **Azure CLI** to provision VMs, storage, and networking resources.
- Developed and maintained **CI/CD pipelines** with **Jenkins** for seamless integration, automated testing, and deployments to AKS clusters.

- Containerized applications using **Docker** and managed images with **Azure Container Registry (ACR)** for secure and consistent deployments.
- Deployed and managed **AKS clusters**, configuring namespaces, node pools, autoscaling, and Kubernetes probes for application reliability and health management.
- Administered **OpenShift** clusters in hybrid cloud environments for advanced container orchestration beyond Kubernetes.
- Configured Azure networking services including **Load Balancers**, **Application Gateways**, and **Traffic Manager** for optimized traffic distribution and availability.
- Executed migrations of physical and virtual workloads to **Azure VMs** using **Azure Site Recovery (ASR)** for business continuity.
- Built **Azure Blob Storage** pipelines for secure, redundant log and data storage.
- Leveraged **Azure Data Factory** for data migration and transformation from on-premises SQL Server to Azure SQL Database.
- Implemented monitoring and alerting using **Grafana**, **Prometheus**, **ELK Stack**, and **Splunk** to ensure proactive infrastructure management.
- Automated infrastructure and operational workflows using **Python** scripting, including querying **Elasticsearch** and managing Kubernetes containers.
- Managed Azure **Role-Based Access Control (RBAC)** and IAM policies to enforce security and compliance.
- Conducted incident resolution and root cause analysis to maintain SLA compliance and system stability.
- Collaborated with cross-functional teams to align DevOps and development workflows for continuous delivery.
- Documented deployment processes, architecture, and configuration changes for team knowledge sharing.

Environment:

Microsoft Azure (VMs, AKS, ACR, Blob Storage, Traffic Manager, Site Recovery, Data Factory, SQL Database, App Services), Terraform, ARM Templates, Azure CLI, Docker, Kubernetes (AKS), OpenShift, Jenkins, Git, ELK Stack, Splunk, Grafana, Prometheus, PowerShell, Python, Elasticsearch, SQL Server

Systems Administrator

e-Noah Solutions, IND,

Jul 2016 – Nov 2018

Project Description:

Provided infrastructure support and administration for internal systems, including both **Linux and Windows servers**, across development and production environments. Deployed and configured **web servers** and automated system operations using **Bash scripting**. Assisted in implementing **Azure-based environments**, setting the stage for modern DevOps practices. Supported version control, continuous integration, and basic networking services while enhancing operational stability and team collaboration.

Key Responsibilities:

- Administered and maintained **Linux (Red Hat, Ubuntu)** and **Windows Server** environments ensuring system stability, security, and availability.
- Installed, configured, and optimized **Apache**, **Nginx**, and **Tomcat** web servers for development and testing environments.

- Automated routine system tasks including log rotation, disk usage alerts, and backups using **Bash/Shell scripting** to reduce manual effort.
- Managed core network services such as **DNS, DHCP, FTP**, and handled **TCP/IP** configuration and troubleshooting for seamless connectivity.
- Provisioned and supported **Azure Virtual Machines**, assisting development teams with infrastructure setup and basic Azure operations.
- Installed software, performed patch management, and conducted system upgrades to maintain environment consistency and security compliance.
- Supported **MySQL** and **SQL Server** database administration including setup, backup, and user access control.
- Implemented **Git** for version control, introducing source tracking and change management best practices to internal teams.
- Configured and deployed **Jenkins** for basic CI pipeline automation, enabling automated builds and testing for developers.
- Created and maintained technical documentation and runbooks for system configurations, troubleshooting, and deployment procedures.

Environment:

Linux (Red Hat, Ubuntu), Windows Server, Azure (VMs, Blob Storage, IAM), Apache, Nginx, Tomcat, Bash, Shell Scripting, Git, Jenkins, MySQL, SQL Server, DNS, DHCP, FTP, TCP/IP.